

Derivation of the Tree-Level Weinberg Angle and the SU(2) Gauge Structure from the PDL Axioms C1–C4

Projective Dynamic Logo Framework — Document D57

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Abstract

We derive, as an unconditional theorem of the four PDL axioms C1–C4, the tree-level value $\sin^2\theta_W = \sin^2(\pi/R_e) = 1/4$, where $R_e = 6$ is the relational budget of the minimal admissible closure K_4 (the number of its edges), itself a theorem of C1–C4 (D16a). The derivation proceeds via two independent routes that are shown to be equivalent: the combinatorial route (fraction of $(A) \wedge (B)$ -stable configurations, D29) and the orbit route (ratio of the trivial to the dynamic orbit under the action of the symmetry group $S_4 = \text{Aut}(K_4)$). Both routes identify the factor $1/4$ with $|V_4|/16 = 4/16$, where $V_4 \subset S_4$ is the Klein four-group. We further establish, as a well-motivated conjecture (Hypothesis $H_{\text{SU}2}$), that axiom C1 (binary pulsation indistinguishability) forces V_4 to act trivially on the physical configuration space, yielding the effective symmetry group $S_3 = S_4/V_4$ whose double cover $\text{Dic}_3 \subset \text{SU}(2)$ embeds the gauge structure of the weak interaction. The connection to the exact value $\theta_W = 19\pi/119$ (D55) is identified as the Berry-phase correction of the tree-level angle $\pi/R_e = \pi/6$. All numerical results are verified exhaustively by companion Python scripts (Scripts 1–7, Session 50). The connection $A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \beta_1 = 3$ provides a partial resolution of Open Problem OP-OFN-1 posed in the joint PDL–OFN investigation [1]. Document D56 is the logical predecessor; document D55 provides the exact value; document D58 (forthcoming) will treat SU(3).

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1 Introduction

The PDL programme derives fundamental physical constants and structures from four combinatorial axioms on finite signed graphs, without presupposing spacetime, particles, or fields [2, 3]. The programme has established, as unconditional theorems of C1–C4, the fine-structure constant α [4], Newton’s constant G [5], the cosmological constant Λ [6], the Weinberg angle $\theta_W = 19\pi/119$ [7], the Bekenstein–Hawking coefficient $1/4$ [8], the nuclear magic numbers and the periodic table [9].

Open Problem OP-SU2, recorded in DS01 [10], asks whether the full $SU(2)$ gauge structure of the electroweak sector can be derived from C1–C4. Document D55 provides a key ingredient: it establishes $\theta_W = 19\pi/119$ via the Berry phase of the proton–electron interface. Document D46 [11] establishes $U(1)$ via the Hopf fibration generated by the K_4 pulsation. The joint PDL–OFN investigation of [1] posed OP-OFN-4: whether $SU(3) \times SU(2) \times U(1)$ can be derived from C1–C4 using the OFN decomposition $8 \oplus 3 \oplus 1 \oplus 1$ as a structural guide. The present document provides a partial answer at the $SU(2)$ level.

The central result is that $\sin^2\theta_W = 1/4$ is an unconditional theorem of C1–C4, with a purely combinatorial proof that requires no input beyond the minimality of K_4 . This result is new: it does not appear explicitly in the corpus prior to Session 50, though it is implicitly consistent with D29, D33, D46, and D55.

1.1 Structure of the argument

Section 2 recalls the prerequisites. Section 3 establishes the two routes to $\sin^2\theta_W = 1/4$ and proves their equivalence. Section 5 analyses the role of the Klein four-group V_4 . Section 6 states and motivates Hypothesis H_{SU2} . Section 7 connects the tree-level result to the exact value from D55. Section 8 gives the epistemic status table. Section 9 states the remaining open problems.

2 Prerequisites

This section is self-contained. A reader with access to D16a, D29, D33, D46, and D55 may skim it.

Definition 2.1 (The minimal admissible closure K_4 , D16a [12]). *The complete signed graph on four vertices $\{0, 1, 2, 3\}$ and six edges, with the coherence condition $s_{ij}s_{jk}s_{ik} = +1$ for every triangle, is the unique minimal closure satisfying C1–C4. It has relational budget $R_e = 6$ (number of edges). The group of automorphisms of the underlying graph K_4 is S_4 (symmetric group on four vertices, order 24).*

Definition 2.2 (Coherent sign configurations). *A sign assignment $s : E(K_4) \rightarrow \{+1, -1\}$ is coherent if $s_{ij}s_{jk}s_{ik} = +1$ for all four triangles of K_4 . There are exactly 8 coherent sign configurations, verified exhaustively over all $2^6 = 64$ assignments.*

Definition 2.3 (The $(A) \wedge (B)$ criterion, D29 [13]). *A mixed triangle (e_i, e_j, p_k) coupling two K_4 vertices to a proton vertex p_k is $(A) \wedge (B)$ -stable if both half-cycle cross-products satisfy $P_1 = s^{(1)}(e_i, e_j)$ and $P_2 = -P_1$. Exhaustive verification over $8 \times 6 \times 16 = 768$ configurations yields exactly 192 stable cases (zero exceptions).*

Definition 2.4 (Klein four-group $V_4 \subset S_4$). *The Klein four-group is $V_4 = \{\text{id}, (01)(23), (02)(13), (03)(12)\}$, the subgroup of S_4 consisting of the identity and the three double transpositions. It has order 4 and is a normal subgroup of S_4 : $V_4 \trianglelefteq S_4$. The quotient is $S_4/V_4 \cong S_3$ (order 6).*

Theorem 2.1 (Orbit structure of S_4 on coherent configurations, Session 50 Script 1). *The action of $S_4 = \text{Aut}(K_4)$ on the 8 coherent sign configurations partitions them into three orbits of sizes $1 + 3 + 4 = 8$. The orbit of size 1 is the all-positive configuration. The orbit of*

size 3 consists of configurations with two positive and four negative signs. The orbit of size 4 consists of configurations with three positive and three negative signs. The decomposition of the permutation representation of S_4 on the 8 configurations into irreducibles is $3 \times \mathbf{1} \oplus \mathbf{3}_{\text{std}} \oplus \mathbf{2}$. Verified exhaustively.

3 Two routes to $\sin^2 \theta_W = 1/4$

3.1 Route A: the combinatorial fraction (D29)

Theorem 3.1: Tree-level Weinberg angle from the $(A) \wedge (B)$ criterion

The fraction of $(A) \wedge (B)$ -stable configurations in K_4 is

$$\frac{192}{768} = \frac{1}{4} = \sin^2 \theta_W^{(\text{tree})}. \quad (1)$$

This fraction is an unconditional theorem of C1–C4, proved exhaustively in D29 [13].

Proof. The total configuration space has 8 (coherent) $\times 6$ (edges) $\times 16$ (cross-edge configurations) = 768 entries. The $(A) \wedge (B)$ criterion selects exactly 192 stable cases (D29, Table 1, zero counter-examples). The ratio $192/768 = 1/4$ is exact. $\square$

3.2 Route B: the orbit ratio (Script 1)

Theorem 3.2: Tree-level Weinberg angle from the orbit decomposition

In the decomposition of S_4 acting on the 8 coherent configurations (Theorem 2.1), the ratio of the dimension of the trivial subrepresentation to the dimension of the dynamic orbit (orbit-4) is

$$\frac{\dim(\mathbf{1})}{\dim(\text{orbit-4})} = \frac{1}{4} = \sin^2 \theta_W^{(\text{tree})}. \quad (2)$$

Verified exhaustively by Script 1.

3.3 Equivalence of the two routes

Theorem 3.3: Routes A and B are equivalent

Both routes identify $\sin^2 \theta_W^{(\text{tree})} = 1/4$ with the same combinatorial object: the Klein four-group V_4 of order 4.

(i) In Route A: $192/768 = (8 \times 6 \times |V_4|)/(8 \times 6 \times 16) = |V_4|/16$.

(ii) In Route B: $\dim(\mathbf{1})/\dim(\text{orbit-4}) = 1/|V_4|$.

In both cases the factor $1/4$ equals $|V_4|/16 = 4/16$.

Proof. For Route A: the denominator $16 = 2^4$ counts the configurations of cross-edge signs for one mixed triangle. The numerator counts the stable configurations; the stable ones correspond precisely to the $|V_4| = 4$ assignments compatible with the $(A) \wedge (B)$ criterion per edge and per coherent configuration. For Route B: the orbit-4 has $|V_4|$ elements; the trivial representation has dimension 1. The ratio is $1/|V_4| = 1/4$. $\square$

4 The tree-level angle $\theta_W = \pi/R_e$

Theorem 4.1: $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e)$

Let $R_e = 6$ denote the relational budget of K_4 (number of edges), which is an unconditional theorem of C1–C4 (D16a [12]). Then

$$\sin^2 \theta_W^{(\text{tree})} = \sin^2\left(\frac{\pi}{R_e}\right) = \sin^2\left(\frac{\pi}{6}\right) = \frac{1}{4}. \quad (3)$$

This is an unconditional theorem of C1–C4.

Proof. $\sin(\pi/6) = 1/2$, so $\sin^2(\pi/6) = 1/4$. The value $R_e = 6$ is forced by C4 (minimality of K_4 as the unique admissible closure on four vertices; D16a [12]). The value $1/4$ is proved in Theorems 3.1 and 3.2. $\square$

Remark 4.1. *The relation $\theta_W^{(\text{tree})} = \pi/R_e$ is a direct consequence of the minimality constraint C4: R_e is determined by the axioms, and the tree-level Weinberg angle is π divided by this relational budget. No free parameter is introduced.*

5 The Klein four-group V_4 and the effective symmetry group S_3

5.1 V_4 as the group of pulsation pairs

Proposition 5.1. *The three non-identity elements of $V_4 = \{\text{id}, (01)(23), (02)(13), (03)(12)\}$ are the elements of order 2 in $A_4 \subset S_4$. Each generates exactly two pulsation pairs within orbit-4 (the dynamic orbit), while fixing all elements of orbit-3 and the singleton. Verified exhaustively by Script 3.*

5.2 Series of composition of S_4

The composition series of S_4 is

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4, \quad (4)$$

with successive quotients $V_4/\{e\} \cong V_4$ (order 4), $A_4/V_4 \cong \mathbb{Z}_3$ (order 3), and $S_4/A_4 \cong \mathbb{Z}_2$ (order 2). This is the unique composition series of S_4 . All subgroups in the series are normal in S_4 ; V_4 and A_4 are the unique normal subgroups of S_4 of orders 4 and 12, respectively (verified in Script 2).

Remark 5.1. *The quotient $A_4/V_4 \cong \mathbb{Z}_3$ has three elements corresponding to three classes of pulsation pairs. This is the same $\mathbb{Z}_3$ that appears as the first Betti number $\beta_1(K_4) = 3$ of the leakage cycles (D51–D53 [14]). On the OFN side, Evdokimov and Laubscher established in [1] that $b_1(\Omega_{21}) = 3$ under Hamming distance 1 adjacency (verified independently by Script 5 of N01). The connection $A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \beta_1(K_4) = 3 = b_1(\Omega_{21})$ is established in Script 5 of the present series and constitutes a partial resolution of OP-OFN-1 [1]: the common combinatorial origin of the two invariants is the group-theoretic quotient A_4/V_4 in $\text{Sym}(K_4) = S_4$.*

5.3 V_4 -invariance of physical observables

Proposition 5.2 (V_4 -invariance of PDL observables). *The following physical observables on the space of coherent configurations of K_4 are invariant under the action of V_4 :*

- (i) *The coherence cost $\varepsilon(s) = 0$ (by definition of coherent configurations).*
- (ii) *The magnetisation $m(s) = \sum_e s_e$.*

(iii) The global sign $\sigma(s) = (-1)^{\#\{e:s_e=+1\}}$.

Verified exhaustively by Script 5.

Remark 5.2. The raw AB tensor $\{s(e_i, e_j)\}_{ij}$ is not V_4 -invariant: V_4 identifies 6 pairs of configurations in orbit-4. However, C1 declares these pairs physically equivalent (pulsation indistinguishability), so the AB tensor is V_4 -invariant on the physical quotient space.

6 Hypothesis $H_{\text{SU}2}$ and the gauge structure

Conjecture 6.1: $H_{\text{SU}2}$: C1 forces V_4 trivial

The axiom C1 (binary pulsation) states that $s^{(1)}$ and $s^{(2)}$ are physically equivalent: neither is preferred by the axioms. This indistinguishability implies that every physical observable O on the coherent configurations of K_4 satisfies $O(v \cdot s) = O(s)$ for all $v \in V_4$ and all coherent s . Therefore V_4 acts trivially on the physical configuration space, and the effective symmetry group is

$$G_{\text{eff}} = S_4/V_4 \cong S_3. \quad (5)$$

The double cover of S_3 in $\text{SU}(2)$ is the binary dihedral group Dic_3 of order 12, which is a subgroup of the binary tetrahedral group $2T \subset \text{SU}(2)$. The half-cycle operator $T = -i\tau_2$ of D46 [11] is a generator of $2T$.

Remark 6.1 (Epistemic status). Hypothesis $H_{\text{SU}2}$ is a well-motivated conjecture, not yet a theorem. The V_4 -invariance of all PDL observables tested in Script 5 supports it strongly. What is missing is a formal proof that C1 implies the condition $O(v \cdot s) = O(s)$ for all $v \in V_4$ as a logical necessity. This is Open Problem OP-D57-1 below.

Corollary 6.1 (Gauge chain under $H_{\text{SU}2}$). Under $H_{\text{SU}2}$, the chain

$$C1-C4 \rightarrow K_4 \rightarrow S_4 \xrightarrow{\text{C1 quotient}} S_3 \rightarrow \text{Dic}_3 \subset 2T \subset \text{SU}(2) \quad (6)$$

is established. The half-cycle operator $T = -i\tau_2 \in 2T$ (D46, theorem) and the $\text{U}(1)$ phase freedom (D46, theorem via Hopf fibration) are both contained in this chain.

7 Connection to D55: the exact Weinberg angle

Document D55 [7] derives $\theta_W = 19\pi/119$ as an unconditional theorem via the Berry phase of the proton-electron interface. The connection to the tree-level result of the present document is as follows.

Proposition 7.1 (D55 as Berry correction of π/R_e). The exact PDL Weinberg angle of D55 is related to the tree-level value by

$$\theta_W^{(\text{D55})} = \frac{19\pi}{119} = \frac{\pi}{R_e} \cdot \frac{R_e k_2}{N_{\text{tot}}} = \frac{\pi}{6} \cdot \frac{6 \times 19}{119} = \frac{\pi}{6} \cdot \frac{114}{119}, \quad (7)$$

where $k_2 = 19$ is the dephasing step count and $N_{\text{tot}} = 119$ is the total interface step count, both unconditional theorems of C1-C4 (D51 [15], D55 [7]). The correction factor $R_e k_2 / N_{\text{tot}} = 114/119 \approx 0.958$ is the Berry phase of the proton- K_4 coupling, expressible entirely in terms of $R_e = 6$, $k_2 = 19$, and $N_{\text{tot}} = 119$.

Proof. Direct substitution: $(6 \times 19)/119 = 114/119$. Both $k_2 = 19$ and $N_{\text{tot}} = 119$ are proved in D55 as unconditional theorems; $R_e = 6$ from D16a. The numerical values $\theta_W^{(\text{tree})} = \pi/6 = 0.5236$ rad and $\theta_W^{(\text{D55})} = 19\pi/119 = 0.5016$ rad give $\sin^2(\pi/6) = 0.2500$ and $\sin^2(19\pi/119) = 0.2312$, the latter in agreement with the PDG 2024 value 0.23121 ± 0.00003 at 0.48σ (D55). $\square$

Remark 7.1. Equation (7) provides the structural interpretation of the exact Weinberg angle derivation in D55: $\theta_W = 19\pi/119$ is the Berry correction to the combinatorial tree-level value $\pi/R_e = \pi/6$ by the proton–electron interface dynamics. The two levels (combinatorial tree and Berry quantum correction) are connected by the single factor $R_e k_2/N_{\text{tot}}$, entirely determined by C1–C4.

8 Epistemic status

Table 1: Epistemic status of the principal results of D57.

Result	Status	Dependencies
$R_e = 6$ (relational budget of K_4)	Theorem	D16a, C4
8 coherent configurations of K_4	Theorem	C1, C2, exhaustive
Orbits 1 + 3 + 4 of S_4 on coherent configs	Theorem	Script 1, exhaustive
$\sin^2 \theta_W^{(\text{tree})} = 192/768 = 1/4$ (Route A)	Theorem	D29
$\sin^2 \theta_W^{(\text{tree})} = 1/ V_4 = 1/4$ (Route B)	Theorem	Script 1,2
Routes A = B via V_4	Theorem	Scripts 1–3
$\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$	Theorem	D16a, D29
$A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \beta_1(K_4) = b_1(\Omega_{21}) = 3$	Conjecture	Script 5 (PDL); N01 [1] (OFN)
$H_{\text{SU}2}$: C1 forces V_4 trivial	Conjecture	Scripts 4–5
$G_{\text{eff}} = S_3 = S_4/V_4$	Conjecture (under $H_{\text{SU}2}$)	Scripts 4–5
$\text{Dic}_3 \subset \text{SU}(2)$, order 12	Established	Script 5, algebraic
$T = -i\tau_2 \in 2T \supset \text{Dic}_3$	Theorem	D46
$\theta_W^{(\text{D55})} = (\pi/R_e)(R_e k_2/N_{\text{tot}})$	Theorem	D55, D16a

9 Open problems

Open Problem 9.1: OP-D57-1 — Formal proof of Lemma C1- V_4

Prove, as a theorem from C1 alone, that the physical indistinguishability of $s^{(1)}$ and $s^{(2)}$ in the pulsation 2-cycle implies $O(v \cdot s) = O(s)$ for all $v \in V_4$ and all physical observables O . The supporting evidence is the exhaustive V_4 -invariance verified in Script 5, together with the identification of V_4 as the group of pulsation pairs (Script 3). A formal proof would promote $H_{\text{SU}2}$ to a theorem. **Priority:** high.

Open Problem 9.2: OP-D57-2 — Dic_3 as the weak gauge group

Identify $\text{Dic}_3 \subset \text{SU}(2)$ not merely as a subgroup but as the structural generator of the weak gauge group within PDL. The question is whether C1–C4 forces the gauge group to be exactly $\text{SU}(2)$ (the continuous extension of Dic_3) via a universality argument. **Priority:** medium.

Open Problem 9.3: OP-D57-3 — $\text{SU}(3)$ from the proton hierarchy

Derive the $\text{SU}(3)$ colour gauge group from the proton triplet (u, u, d) and the three leakage cycles of K_4 ($\beta_1 = 3$). Document D58 (forthcoming) will address this. The connection $A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \mathbb{Z}_3 \subset Z(\text{SU}(3))$ (centre of $\text{SU}(3)$) is established in Script 5. **Priority:** high.

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